Customer Shopping Behavior Analysis

This project analyzes customer shopping behaviorusing transactional data

from 3,900 purchases. Ourgoal istouncover insightsinto spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.

Project Overview

Transactional Data

Analysis of3,900 purchasesacross various product categories.

Uncover Insights

Identifyspendingpatterns, customer segments, and product preferences.

Guide Decisions

Informstrategicbusiness decisions and optimize marketing efforts.

Dataset Summary

Rows: 3,900

Columns: 18

Missing Data: 37 values in 'Review Rating'

Key Features:

- Customer demographics (Age, Gender, Location, Subscription Status)
- Purchase details (Item, Category, Amount, Season, Size, Color)
- Shopping behavior (Discount, Promo Code, Previous Purchases, Frequency, Review, Shipping)

Exploratory Data Analysis (Python)

01

Data Loading & Exploration

Imported datasetwith `pandas`;checked structure and summary statistics.

02

Missing Data Handling

Imputed missing 'Review Rating'values using the median rating per product category.

03

Column Standardization

Renamed columns to `snake_case` for readability.

04

Feature Engineering

Created `age_group` and `purchase_frequency_days` columns.

05

Database Integration

LoadedcleaneddataintoPostgreSQL for SQL analysis.

Data Analysis using SQL

Revenue by Gender

Female: $75,191

Male: $157,890

High-Spending Discount Users

Identified839customers who used discounts but spent above average.

Top 5 Products by Rating

Gloves(3.86),Sandals(3.84),Boots(3.82), Hat (3.80), Skirt (3.78).

SQL Insights: Shipping & Subscriptions

Shipping Type Comparison

Standard: $58.46 average purchase

Express: $60.48 average purchase

Subscribers vs.Non-Subscribers

Subscribers(1,053):Avg Spend$59.49,Total Revenue $62,645

Non-Subscribers (2,847): Avg Spend $59.87, Total Revenue $170,436

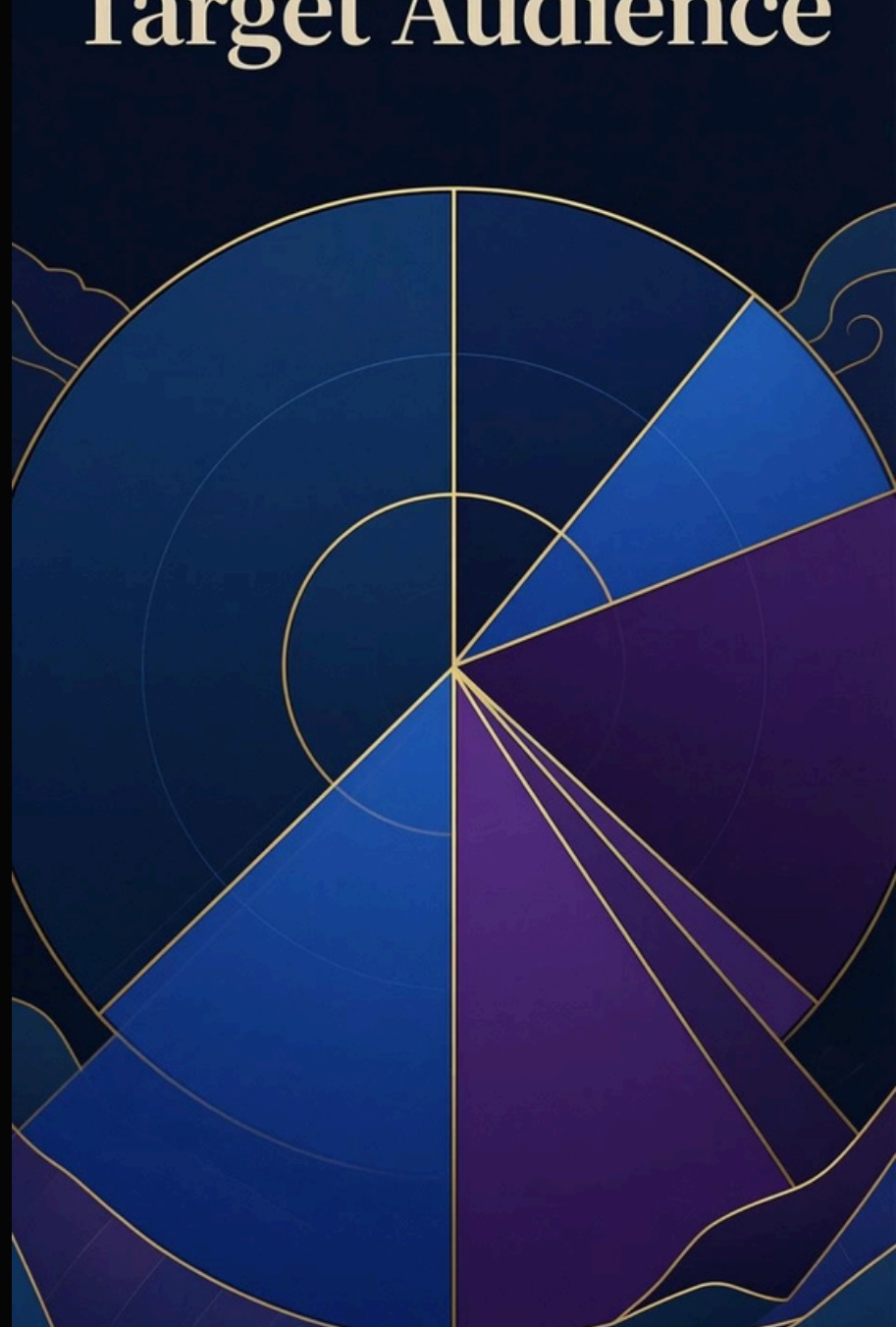
SQL Insights: Products & Customers

Discount-Dependent Products

Hat(50%),Sneakers(49.66%),Coat(49.07%), Sweater (48.17%), Pants (47.37%).

Customer Segmentation

Loyal (3,116), Returning (701), New (83).



SQL Insights: Top Products & Age Groups

Top 3 Products per Category

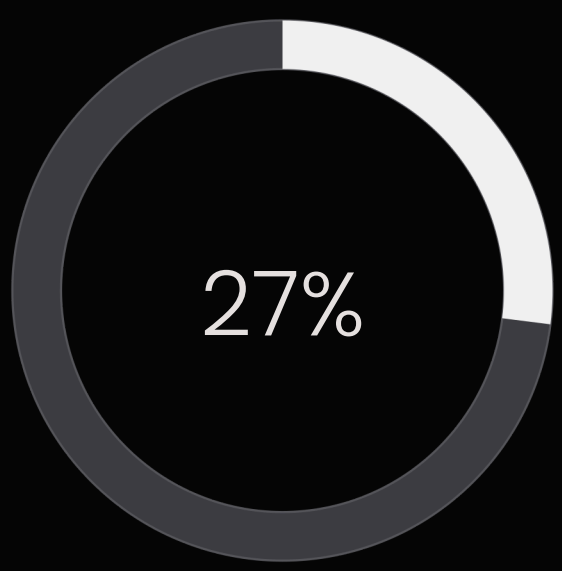
- **Accessories:** Jewelry, Sunglasses, Belt
- **Clothing:** Blouse, Pants, Shirt
- **Footwear:** Sandals, Shoes, Sneakers
- **Outerwear:** Jacket, Coat

Revenue by Age Group

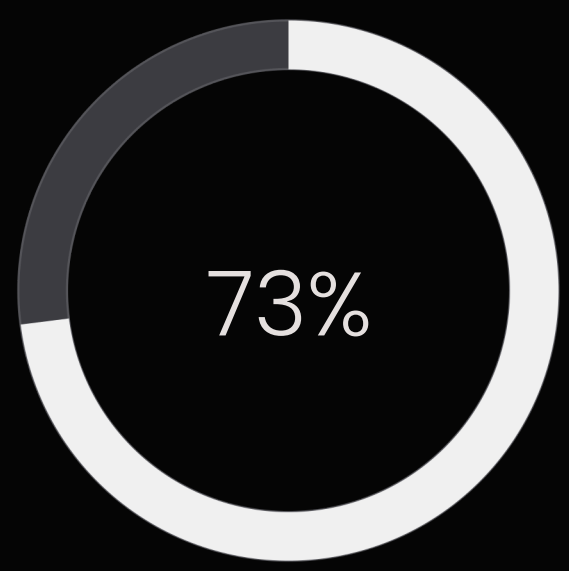
- **Young Adult:** $62,143
- **Middle-aged:** $59,197
- **Adult:** $55,978
- **Senior:** $55,763

Dashboard in Power BI

An interactive Power BI dashboard was built to visually present key insights.



Subscribers



Non-Subscribers

Business Recommendations

BoostSubscriptions

Promoteexclusivebenefitsfor subscribers.

Customer Loyalty Programs

Reward repeat buyers to foster loyalty.

Review Discount Policy

Balance sales boosts with margin control.

Targeted Marketing

Focuseffortsonhigh-revenue age groups and express-shipping users.